THERMOCHEMISTRY

→ It deals with the measurement of heat of reaction under given set of conditions using calorimeter.

* Heat of Reaction :-

It is heat released or absorbed when 1 molar quantities of reactants indicated by Stoichiometry of given reaction have completely reacted to form products.

eg. $2A + 3B \longrightarrow C + 2D$ (Balanced rkn.)

2 mol 3 mol 1 mol 2 mol Heat of rkn = 100 KJ/mol

Per mol means per mol extent of rkn.

→ Unit of Heat of Reaction is KJ/mol or Kcal/mol.

* Exothermic reaction → Heat liberates.
* Endothermic reaction → Heat is absorbed.

eg. $A + B \longrightarrow C, \quad \Delta H_r = -50\ KJ/mol$

OR

$A + B \longrightarrow C + 50\ KJ$ (Exothermic)

eg. $X + 2Y \longrightarrow Z, \quad \Delta H_r = 200\ KJ/mol$

OR

$X + 2Y + 200\ KJ \longrightarrow Z$ (Endothermic)

* Factors affecting Heat of Reaction :-

(1) At constant volume or At constant Pressure :

$Q_v = \Delta U$

$Q_p = \Delta H$

→ Experimentally, measured in Bomb Calorimeter or constant volume Calirometer.

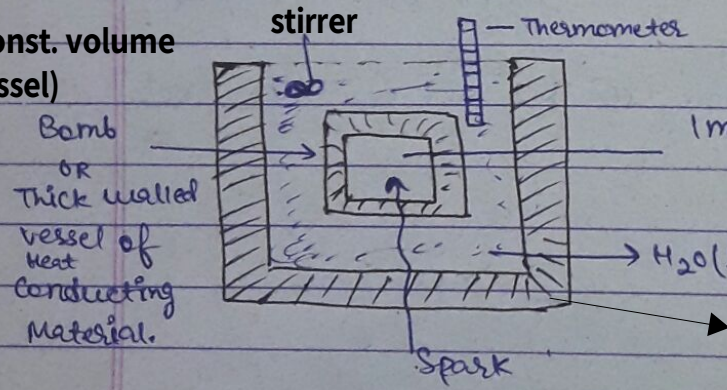
↳ Experimentally it is measured in constard pressure Calorimeter.

$$\Delta H_r = \Delta U_r + \Delta n_g \times RT$$

$$\Delta H_r = \Delta U_r + \Delta(PV)$$

* Bomb Calorimeter : eg. Combustion of CH_4

(const. volume vessel)



thermal insulation to minimize loss of heat

Temperature of water $t_1 \rightarrow t_2$

$$Q_v = \Delta U = m_{water} \times (\text{Specific Heat})_{water} \times (t_2 - t_1) + m_2 s_2 \Delta t + m_3 s_3 \Delta t + \ldots\ldots$$

* water Equivalent (W.E.) → mass of water that would absorb the same heat and actually absorbed by Stirrer, bomb, thermometer etc.

$$\Delta U = Q_v = (m_{water} + W.E) \times s \times \Delta t$$

(2) Physical state or allotropic form :-

eg. $H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(\ell)$, $\Delta H_r = x$ kJ/mol

$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(g)$, $\Delta H_r = y$ kJ/mol

$x \neq y$

eg. $C(dia) + O_2(g) \longrightarrow CO_2$, $\Delta H_r = a$ kJ/mol

$C(graph) + O_2(g) \longrightarrow CO_2$, $\Delta H_r = b$ kJ/mol, $a \neq b$

(3) Effect of temperature :- (Kirchoff's equation)

$aA + bB \longrightarrow cC + dD$

Tem, T_1 aH_A bH_B cH_C dH_D molar enthalpies at temp. T1

$$(\Delta H_r)_{T_1} = \sum (H)_{prod} - \sum (H)_{React.}$$

$$= (cH_C + dH_D) - (aH_A + bH_B)$$

Temp. T_2 $\left\{aH_A + \int_{T_1}^{T_2} aC_p(A)\,dT\right\}$, $\left\{bH_B + \int_{T_1}^{T_2} bC_p(B)\,dT\right\}$ ....

$$(\Delta H_r)_{T_2} = (\Delta H_r)_{T_1} + \int_{T_1}^{T_2} \{cC_p(C) + dC_p(D) - aC_p(A) - bC_p(B)\}\,dT$$

$$\Rightarrow (\Delta H_r)_{T_2} - (\Delta H_r)_{T_1} = \int_{T_1}^{T_2} (\Delta C_p)_r\,dT$$

$$\Rightarrow \boxed{\frac{(\Delta H_r)_{T_2} - (\Delta H_r)_{T_1}}{(T_2 - T_1)} = (\Delta C_p)_r}$$ Kirchoff's eq^n.

where $(\Delta C_p)_r = \Sigma (C_p)_{prod.} - \Sigma (C_p)_{Reactant}$

Sly

$$\frac{(\Delta U_r)_{T_2} - (\Delta U_r)_{T_1}}{(T_2 - T_1)} = (\Delta C_v)_r$$

(*) Various Types of Heat of Reaction :-

1. Standard Enthalpy of formation (ΔH_f°):
It is the enthalpy change when 1 mol of a substance is formed from its elements in their standard state.

- Definition of Standard State
  - → For any gas : Ideal and pressure = 1 bar
  - → For aq. solution : Molarity = 1 Mol/lit at 1 bar pressure.
  - → For solids and liquids : Pure at 1 bar pressure.

Sme → Standard State of an element is generally the most stable state of element at 1 bar pressure and at temperature of interest.
(usually at 25°C or 298.15K) = ambient temperature

* Temperature is not the part of standard state.

Standard State of elements at 25°C :-

Elements		Standard State
Hydrogen	:	$H_2(g)$
Iron	:	Fe(s)
Bromine	:	$Br_2(l)$
Iodine	:	$I_2(s)$
Sulphur (rh.)	:	S (rhombic)
Carbon	:	C (graphite), s
* Phosphorous	:	white phosphorous

P(white) < P(red) < P(Black)

→ increasing thermodynamic stability.

* Though P(white) is least stable, yet it is chosen as standard state of phosphorus.
'P' white is easily reproducable form.

Tin : Sn (white)

oxygen : $O_2(g)$

ΔH_f° of all the elements in standard state is assigned zero.

$\Delta H_f^\circ (C_{(graph)}) = 0$, $\Delta H_f^\circ (C(dia)) \neq 0$

$\Delta H_f^\circ (P(white)) = 0$, $\Delta H_f^\circ (P(Black)) \neq 0$ infact, it is -ve

$\Delta H_f^\circ (Br_2(l)) = 0$; $\Delta H_f^\circ (Br_2(g)) \neq 0$

$\Delta H_f^\circ (H^+(aq)) = 0$ Remember!

imp.

eg. $\frac{1}{2}H_2(g) + \frac{1}{2}Cl_2(g) \longrightarrow HCl(g)$, $\Delta H_r^\circ = \Delta H_f^\circ (HCl_{(g)})$

free radical and anion-cation are not standard states

$H^+(g) + Cl^-(g) \longrightarrow HCl(g)$, $\Delta H_r^\circ \neq \Delta H_f^\circ (HCl)$

$H(g) + Cl(g) \longrightarrow HCl(g)$, $\Delta H_r^\circ \neq \Delta H_f^\circ (HCl)$

$H_2(g) + Cl_2(g) \longrightarrow 2HCl(g)$, $\Delta H_r^\circ \neq \Delta H_f^\circ (HCl)$

as 2 moles are formed

eg $\frac{1}{2}N_2(g) + 2H_2(g) + \frac{1}{2}Cl_2(g) \longrightarrow NH_4Cl(s)$, $\Delta H_r^\circ = \Delta H_f^\circ (NH_4Cl)$

eg $Mg(s) + Cl_2(g) \longrightarrow MgCl_2(s)$, $\Delta H_r^\circ = \Delta H_f^\circ (MgCl_2(s))$

(*) Application of ΔH_f° :

1. Higher is ΔH_f° of a substance, lower will be its thermodynamic stability.
(used in comparing stability of among isomers or allotropes)

Q. which has/have -ve ΔH_f° -

A. C(dia) C. P(red)

B. $Br_2(g)$ D. P(Black)

2. To calculate Enthalpy of Reaction (ΔH_r°)

$aA + bB \longrightarrow cC + dD$.

$$\Delta H_r^\circ = \sum (\Delta H_f^\circ)_{prod.} - \sum (\Delta H_f^\circ)_{Reactants}$$
$$= c \times \Delta H_f^\circ (C) + d \times \Delta H_f^\circ (D) - a \times \Delta H_f^\circ (A) - b \times \Delta H_f^\circ (B)$$

Q Calculate ΔH_f° of $CH_3OH(l)$ at 300K, if

$$CH_3OH(l) + \frac{3}{2}O_2(g) \longrightarrow CO_2(g) + 2H_2O(l), \quad -715 \text{ KJ/mol} \quad \text{— (i)}$$

$$C(\text{graph}) + O_2(g) \longrightarrow CO_2(g), \quad -393 \text{ KJ/mol} \quad \text{— (ii)}$$

$$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l), \quad -285 \text{ KJ/mol} \quad \text{— (iii)}$$

Soln:

$$-715 = \sum (\Delta H_f^\circ)_{prod} - \sum (\Delta H_f^\circ)_{reactants}$$

$$-715 = \left\{1 \times (-393) + 2 \times (-285)\right\} - \left\{x + \frac{3}{2} \times 0\right\}$$

$$-715 = -393 + (-570) - x$$

$$x = 715 - 963$$

$$\boxed{x = -248 \text{ KJ/mol.}}$$

* Alternate Method :-

Desired reaction,

$$C(\text{graph}) + 2H_2(g) + \frac{1}{2}O_2(g) \longrightarrow CH_3OH(l), \quad \Delta H_f^\circ$$

$$\left\{(ii) + 2 \times (iii) - (i)\right\}$$

$$\Delta H_f^\circ = -393 + 2 \times (-285)$$

$$\Delta H_f^\circ = -393 + 2 \times (-285) - (-715)$$

$$\boxed{\Delta H_f^\circ = -248 \text{ KJ/mol}}$$

(2) Enthalpy of Combustion (ΔH_{comb}):

It is the enthalpy change when 1 mol of a substance (a fuel) is burned in excess of oxygen.

eg. $C(s) + O_2(g) \rightarrow CO_2(g)$, $\Delta H_r = \Delta H_{comb}(C(s))$

$CH_4(g) + O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$,

$\Delta H_r = \Delta H_{comb}(CH_4)$

$2Al(s) + \frac{3}{2}O_2(g) \rightarrow Al_2O_3(s)$, $\Delta H_r \neq \Delta H_{comb}(Al(s))$

$Al(s) + \frac{3}{4}O_2(g) \rightarrow \frac{1}{2}Al_2O_3(s)$, $\Delta H_r = \Delta H_{comb}(Al(s))$

V.imp

* For solid elements, 1 mol of atom be considered for ΔH_{comb}.

$S(s) + O_2(g) \rightarrow SO_2(g)$, $\Delta H_r = H_{comb}(S(s))$

$S_8(s) + 8O_2(g) \rightarrow 8SO_2(g)$, $\Delta H_r \neq \Delta H_{comb}(S(s))$

Note:

$C_xH_yO_zS_mN_p\ldots + O_2(g)$
(organic compound)

$\rightarrow CO_2(g) + H_2O(l) + SO_2(g) + N_2\uparrow$

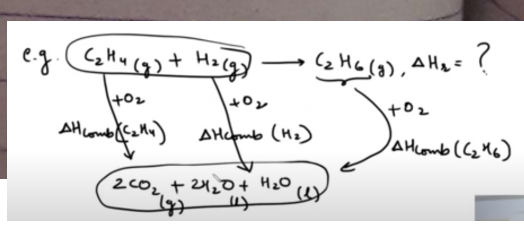
* Application of ΔH_{comb}:

(1.) To determine ΔH_r:

eg: $C_2H_4(g) + H_2(g) \longrightarrow C_2H_6(g)$, $\Delta H_r = ?$

$\Delta H_{comb}(C_2H_4)$ ↓ $+O_2$; ↓ $+O_2$; $+O_2$ →

$2CO_2(g) + 2H_2O(l) + 3H_2O(l)$



- Making a parallel or alternate path of a process or reaction is called Born-Haber cycle.

- Change in State function is independent of path followed. Change in enthalpy, entropy, Gibb's energy (any state function) is independent of path followed.
  { Hess's law }

$$\Delta H_{comb}(C_2H_4) + \Delta H_{comb}(H_2) = \Delta H_r + \Delta H_{comb}(C_2H_6)$$

$$\Rightarrow \boxed{\Delta H_r = \sum(\Delta H_{comb})_{Reactant} - \sum(\Delta H_{comb})_{product}}$$

2nd application:

(2.) Calorific Value (CV)

It is the heat released when 1g of fuel substance is burned in excess of oxygen.

$$\boxed{C.V = \frac{\Delta H_{comb}}{\text{Molar Mass}}}$$

→ Among conventional fuel, calorific value of H_2 is highest.

→ Among hydrocarbons, $\frac{H}{C}$ molar Ratio ↑, CV ↑

eg. $CH_4 > C_2H_6 > C_2H_4 > C_2H_2$

$\left(\frac{H}{C}\right)$ $\frac{4}{1}$ $\frac{3}{1}$ $\frac{2}{1}$ $\frac{1}{1}$

——————→ decreasing C.V

Q. $C_3H_8(g) + H_2(g) \longrightarrow C_2H_6(g) + CH_4(g)$, $\Delta H_r^\circ = ?$

$(\Delta H_{comb}) \rightarrow$ $H_2(g) = -285$ KJ/mol
$CH_4(g) = -890$ KJ/mol
$C_2H_6(g) = -1560$ KJ/mol
$C(graphite) = -393$ KJ/mol

$\Delta H_f^\circ(C_3H_8(g)) = -104$ KJ/mol

Sol^n: $\Delta H_r^\circ = \Delta H_{com}(C_3H_8) + (-285) - (-1560) - (-890)$ — (i)

$3C(graph) + 4H_2(g) \longrightarrow C_3H_8$, -104 KJ

$-104 = 3\times(-393) + 4\times(-285) - \Delta H_{comb}(C_3H_8)$ — (ii)

Putting $\Delta H_{comb}(C_3H_8)$ value in eq (i)

$$\Delta H_r^\circ = -50 \text{ KJ/mol}$$

(3) Enthalpy of Atomisation : (ΔH_{atom}) :

It is the enthalpy change when 1 mol of a substance is converted into its constituent gaseous atoms. (Endothermic process)

eg. $C(s) \longrightarrow C(g)$, $\Delta H_r = \Delta H_{atom}(C(s))$

eg. $Br_2(l) \longrightarrow 2Br(g)$, $\Delta H_r = \Delta H_{atom}(Br_2(l))$

eg. $CH_4(g) \longrightarrow C(g) + 4H(g)$, $\Delta H_r = \Delta H_{atom}(CH_4)$

(4) Lattice Energy / Enthalpy :- ($\Delta H_{L.E}$)

It is the enthalpy change when 1 mol of ionic lattice of a compound is converted into the constituent gaseous ions.

eg. $NaCl(s) \longrightarrow Na^+(g) + Cl^-(g)$, $\Delta H_r = \Delta H_{L.E}(NaCl)$

eg. $MgCl_2(s) \longrightarrow Mg^{2+}(g) + 2Cl^-(g)$, $\Delta H_r = \Delta H_{L.E}(MgCl_2(s))$

$L.E$ or $\Delta H.E \propto \frac{q_1 q_2}{r}$

↑ Potential Energy

$Na^+ Cl^-$ (ionic lattice)

$q_1 \times q_2 = (1)$

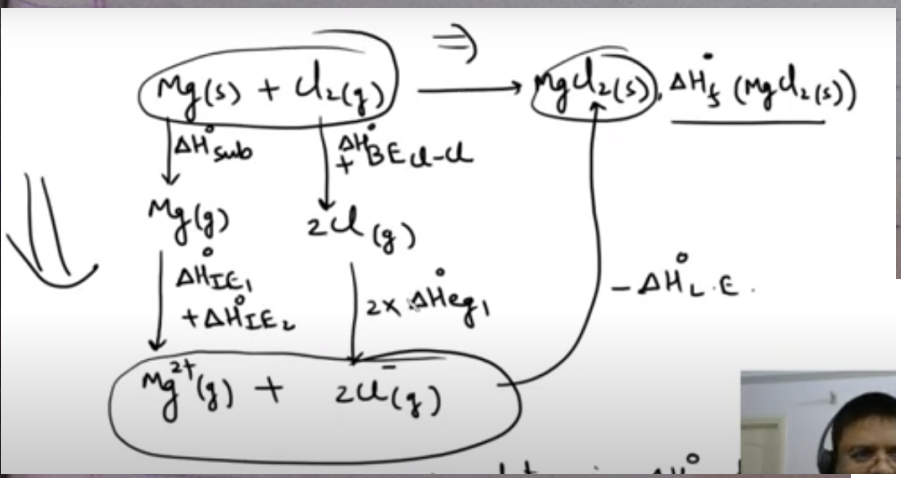
$LiCl > NaCl > KCl > RbCl > CsCl$

Decreasing $L.E$

$MgO > MgS$

→ We can find $\Delta H^\circ_{L.E}$ if ΔH°_f of ionic solid is known and vice-verse

$Mg(s) + Cl_2(g) \longrightarrow MgCl_2(s)$, $\Delta H^\circ_f (MgCl_2(s))$



Born-Haber cycle to determine ΔH°_f of $MgCl_{2(s)}$

Hess's law :

$\Delta H^\circ_f (MgCl_2) = \Delta H^\circ_{sub} + \Delta H^\circ_{IE_1} + \Delta H^\circ_{IE_2} + \Delta H_{BE}(Cl-Cl) + 2\times\Delta H^\circ_{eg}(Cl) + (-\Delta H^\circ_{L.E}(MgCl_2))$

(5) Enthalpy of Phase Transformation :-

a. Enthalpy of fusion (ΔH_{fus}):
It is the enthalpy change when 1 mol of solid is converted into its liquid form. (without any chemical change)

$$H_2O(s) \longrightarrow H_2O(l), \quad \Delta H_r = \Delta H_{fus}(H_2O(s))$$
(Endothermic)

b. Enthalpy of vapourisation (ΔH_{vap})
It is the enthalpy change when 1 mol of a liquid is converted into gas form.

$$H_2O(l) \longrightarrow H_2O(g), \quad \Delta H_r = \Delta H_{vap}(H_2O(l))$$
(Endothermic)

c. Enthalpy of Sublimation :- (ΔH_{sub})
It is the enthalpy change when 1 mol of solid is converted into gas form.

$$H_2O(s) \longrightarrow H_2O(g), \quad \Delta H_r = \Delta H_{sub}(H_2O(s))$$

Note :- For a given substance,

$$\Delta H_{sub} = \Delta H_{fusion} + \Delta H_{vap.}$$

↳ applicable when all enthalpy change data are taken at same temp.

→ To convert heat of reaction at the desired temperature, use Kirchoff's equation.

$$\frac{(\Delta H_r)_{T_2} - (\Delta H_r)_{T_1}}{(T_2 - T_1)} = (\Delta C_p)_r$$

⑥ Bond Enthalpy :-

(a). Bond dissociation enthalpy (ΔH_{BDE})

It is the enthalpy change when 1 mol of a bond from a gaseous reactant (substance) is broken to form gaseous species (free radicals)

eg. $H_2(g) \longrightarrow 2H(g)$, $\Delta H_r = \Delta H_{BDE}(H_2)$

eg. $Br_2(g) \longrightarrow 2Br(g)$, $\Delta H_r = \Delta H_{BDE}(Br_2(g))$

eg. $CH_4(g) \longrightarrow CH_3(g) + H(g)$, $\Delta H_r = \Delta H_{BDE_1} = x$

$CH_3(g) \longrightarrow CH_2(g) + H(g)$; $\Delta H_2 = \Delta H_{BDE_2} = y$

$CH_2(g) \longrightarrow CH(g) + H(g)$, $\Delta H_r = \Delta H_{BDE_3} = z$

$CH(g) \longrightarrow C(g) + H(g)$, $\Delta H_r = \Delta H_{BDE_4} = p$

(b) Average Bond Enthalpy or Bond Enthalpy or Bond Energy. ($\Delta H_{B.E}$)

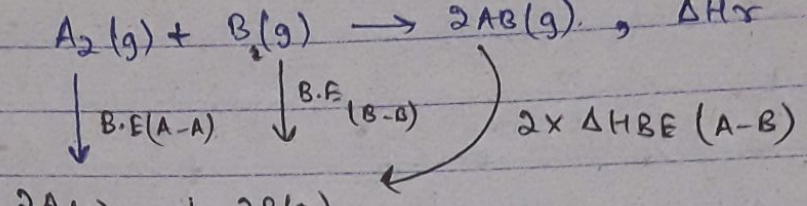
It is the average energy required to break 1 mol of bond between two atom from various gaseous reactants into gaseous atoms.

$$\Delta H_{B.E} = \frac{\Delta H_{BDE_1} + \Delta H_{BDE_2} + \Delta H_{BDE_3} + \ldots \Delta H_{BDE_n}}{n}$$

→ In numerical problems, we use average Bond energy.

* Application of Bond Energy :

1. To determine Enthalpy of Reaction (ΔH_r) :



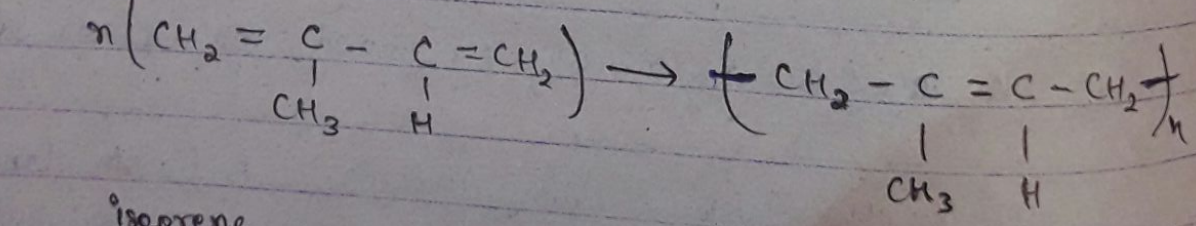
Applying Hess's law,

$$\Delta H_r + 2 \times BE_{A-B} = B.E_{(A-A)} + B.E_{(B-B)}$$

$$\Rightarrow \boxed{\Delta H_r = \sum (B.E)_{Reactants} - \sum (B.E)_{products}}$$

Applicable when all the reactants and products are in gaseous form.

Q. Calculate enthalpy of polymerisation per mol of isoprene.

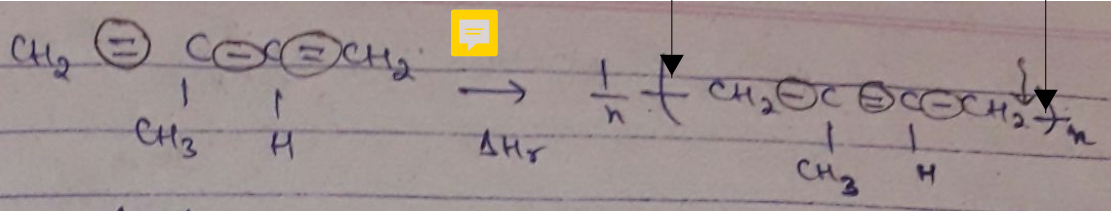


isoprene → Neoprene.

Given, $B.E_{(C=C)} = 590$, $B.E_{(C-C)} = 331$ KJ/mol

only take 1/2 bond for these two bonds (as the other half will go into the other mole, these bonds are shared b/w two moles)

Soln:



1 mol

$$\Delta H_r = \Sigma (B.E)_R - \Sigma (B.E)_P$$

$$= (2 \times B.E_{(C-C)} + B.E_{(C-C)}) - (B.E_{C=C} + 3 \times B.E_{C-C})$$

$$= 590 - 2 \times 331$$

$$= 590 - 662$$

$$= -72 \text{ KJ/mol}$$

Q. Find average P-Cl bond energy in PCl_5

$\Delta H_f^\circ \rightarrow PCl_5(g) = -375$, $P(g) = 315$, $Cl(g) = 122$ KJ/mol.

Soln: 1st method:

$$P(g) + 5Cl(g) \longrightarrow PCl_5(g), \Delta H_r$$

$$\Delta H_r = \Sigma (\Delta H_f^\circ)_P - \Sigma (\Delta H_f^\circ)_{React} = \Sigma (B.E)_R - \Sigma (B.E)_P$$

$$(-375 - 315 - 5 \times 122) = 0 + 5 \times 0 - 5 \times B.E_{P-Cl}$$

$$\Rightarrow \boxed{B.E_{P-Cl} = 260 \text{ KJ/mol}}$$

2nd Approach

$$P(s) + \frac{5}{2} Cl_2(g) \longrightarrow PCl_5(g), -375$$

315 ↓ , $\frac{5}{2}$ → 5×122 ↓

$P(g)$ $5Cl(g)$ → $PCl_5(g)$

$$\Delta H' = \Sigma (B.E)_R - \Sigma (B.E)_P$$

$$= (0 + 5 \times 0) - 5 \times B.E_{P-Cl}$$

$$-375 = 315 + 5 \times 122 + \Delta H'$$

$$\Delta H' = 260 \text{ kJ/mol}$$

(7) Enthalpy of Hydration : (ΔH_{HE})

a. For ions :

It is the enthalpy change when 1 mol of gaseous ion is dissolved in excess of water, to form hydrated ions. (always exothermic)

$Na^+(g) \longrightarrow Na^+(aq)$, $\Delta H_r = \Delta H_{H.E}(Na^+(g))$

1 mol

$Cl^-(g) \longrightarrow Cl^-(aq)$, $\Delta H_r = \Delta H_{H.E}(Cl^-(g))$

(b) For Crystalline Solids :-

It is the enthalpy change when 1 mol of an ionic solid (crystalline solid) is mixed with required amount of water to form partially hydrated or completely hydrated solid.

eg. $CuSO_4(s) + 5H_2O(l) \longrightarrow CuSO_4 \cdot 5H_2O(s)$, ΔH_r = $\Delta H_{H.E}$ ($CuSO_4$)

white crystal — Blue crystals.

eg. $CaCl_2(s) + 2H_2O(l) \longrightarrow CaCl_2 \cdot 2H_2O(s)$, $\Delta H_r = \Delta H_{H.E}$ ($CaCl_2$)

hydrated solid.

(8) Enthalpy of Solution (ΔH_{sol}):

It is the enthalpy change when 1 mol of a compound (may be solid or liquid or gas) is dissolved in excess of water to form aqueous solution.

(Endothermic or Exothermic).

eg. $HCl(g) \longrightarrow H^+(aq) + Cl^-(aq)$, $\Delta H_r = \Delta H_{sol}(HCl(g))$

eg. $CuSO_4(s) \longrightarrow Cu^{2+}(aq) + SO_4^{2-}(aq)$, $\Delta H_r = \Delta H_{sol}(CuSO_4)$

eg. $CaCl_2(s) \longrightarrow Ca^{2+}(aq) + 2Cl^-(aq)$, $\Delta H_r = \Delta H_{sol}(CaCl_2)$

⇒ Integral Enthalpy of Solution (ΔH_{IS} / IES)

It is the enthalpy change when 1 mol of a compound is mixed into required amount of water to form solution of desired concentration.

eg. $HCl(g) + 10H_2O(l) \longrightarrow HCl(aq)\cdot(10H_2O)$, $\Delta H_r = \Delta H_{IS} = x$

$HCl(g) + 20H_2O(l) \longrightarrow HCl(aq)(\cdot 20H_2O)$, $\Delta H_r = \Delta H_{IS} = y$

$HCl(g) + 25H_2O(l) \longrightarrow HCl(aq)(\cdot 25H_2O)$, $\Delta H_r = \Delta H_{IS} = z$

$HCl(g) + \text{Excess of } H_2O(l) \longrightarrow H^+(aq) + Cl^-(aq)$, $\Delta H_r = \Delta H_{sol}(HCl)$

Such solution where heat absorption / liberation stops on further addition of water is called infinitely dilute solution.

* Enthalpy of dilution ($\Delta H_{dil.}$)

It is the enthalpy change when solution of 1 mol of solute is diluted from higher concentration to lower concentration.

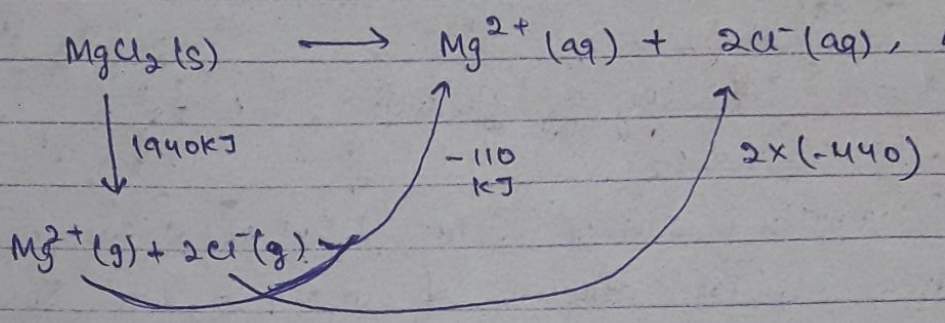
eg. $HCl(aq)\cdot(10H_2O) \longrightarrow HCl(aq)(\cdot 20H_2O)$, $\Delta H_r = \Delta H_{(dil)}$

$(HCl.(10H2O) \longrightarrow (HCl.(20H2O)$

Q. Find enthalpy of solution of $MgCl_2(s)$, if

Hydration of energy of $Mg^{2+}(g)$ = -110 kJ/mol
Hydration of energy of $Cl^-(aq)$ = -440 kJ/mol
Lattice enthalpy of $MgCl_2(s)$ = 1940 kJ/mol.

Soln:



Hess's law, $\Delta H_{soln} = 1940 + (-1110) + 2 \times (-440)$

$= 1940 - 1990$

$= -50$ kJ/mol

Q. Find ^standard Enthalpy of solution of HCl(g) if

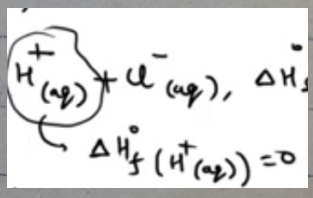
$\Delta H_f^\circ (HCl(g)) = -92.3$ kJ/mol

$\Delta H_f^\circ (Cl^-(aq)) = -167.44$ kJ/mol

Soln: $HCl(g) \longrightarrow H^+(aq) + Cl^-(aq)$, ΔH_{sol}

$\Delta H^\circ_{soln} = 0 + (-167.44) - (-92.3)$

$= -75.14$ kJ/mol.



Enthalpy of neutralisation:

(3.) Enthalpy of neutralisation :- (ΔH_{neut})

It is the enthalpy change when 1 g eq. of an acid is neutralized with 1g eq. of a base to form salt and water. (Always Exothermic)

unit of ΔH_{neut} : kJ/g.eq. or kCal/g.eq.

No. of gram equivalent = No. of moles × valency factor.

→ for complete neutralization,

V.F of HCl = 1	V.F of NaOH = 1
V.F of H_2SO_4 = 2	V.F of $Ba(OH)_2$ = 2.
V.F of H_3PO_4 = 3	

→ ⊛ ΔH_{neut} depends on following two factors.

(i) Nature of acid & Base : (i.e SA or WA & SB or WB)
major factor

(ii) Conc. of acid and Base undergoing neutralization.
(minor factor)
ΔH_{neut} varies little due to change in conc. of acid and base neutralized.

⇒ Neutralization of SA and SB –

$NaOH(aq) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l)$, ΔH_{neut}

0 energy 0 energy 0 0

$Na^+ + (OH^-)_{aq}$ $H^+ + Cl^-_{(aq)}$ $Na^+_{(aq)} + Cl^-_{(aq)}$ $H_2O_{(l)}$
(aq) (aq)

1 mol 1 mol

-13.7 kCal

1 Cal = 4.18 J

$\Delta H_{neut} = -13.7$ kcal / g.eq

$= -57.3$ kJ / g.eq

→ ΔH_{neut} of all SA and SB is nearly constant and is equal to -13.7 kcal / g.eq (or) -57.3 kJ / g.eq

⇒ ΔH_{neut} of WA and SB :-

↑ CH_3COOH ↑ NaOH

$HA(aq) + NaOH(aq) \rightarrow NaA(aq) + H_2O(l), \Delta H_{neut.}$

$\Delta H_{ion.}$ (HA) ↓ ; NaOH ↓ 0 ; NaA (aq) → 0 ; $H_2O(l)$ → 0

$H^+(aq) + A^-(aq)$ + $Na^+(aq) + OH^-(aq) \longrightarrow Na^+ + A^-(aq) + H_2O(l)$

1 mol 1 mol -13.7 kcal (aq)

$\Delta H_{neut} = (\Delta H_{ion}(HA) - 13.7)$ kcal / g.eq.

↑ +ve (endothermic)

⇒ If both acid (HA) and base (BOH) are weak, then

$\Delta H_{neut.} = (\Delta H_{ion}(HA) + \Delta H_{ion}(BOH) - 13.7)$ kcal/g.eq

Q. When 1 mol of aq. NH_3 is neutralized with 1 mol of $HCOOH(aq)$, energy released is 10 kCal, when ammonia is neutralized with $HCl(aq)$, ΔH_{neut} is -12 kCal/g·eq
Find ΔH_{ion} of NH_3 and of $HCOOH$

Sol^n: $\Delta H_{ion}(NH_3) = 1.7$ kcal/mol

$\Delta H_{ion}(HCOOH) = 2$ kcal/mol

Q. Calculate heat of neutralization when 200 ml of 1M $HCl(aq)$ is mixed with 400 ml, 0.5 M $NaOH(aq)$, Temp rise in Calorimeter was found to be 4.4°C. Water equivalent of Calorimeter = 12g.
Density of solution = 1 g/ml.
Specific Heat of Solution = 1 cal/g/k.

Sol^n: $HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$
0.2 mol 0.2 mol

$$q = -\frac{(200+400+12)\times 1\times 4.4}{1000} \text{ Kcal}$$

$$\Delta H_{neut} = -\frac{0.612\times 4.4}{0.2} \text{ Kcal/g·eq.}$$

$$\Delta H_{neut} = -13.464 \text{ kal/g·eq}$$

$$= -\frac{13.464 \text{ KCal/g·eq.}}{\text{or KCal/mol.}}$$

Q → When 1 mol of $H_3PO_{4(aq)}$ is completely neutralized with $NaOH_{(aq)}$, heat released is 150 KJ/mol of H_3PO_4.
Find Enthalpy of ionisation of H_3PO_4.

Sol^n: $H_3PO_{4(aq)} + 3NaOH_{(aq)} \longrightarrow Na_3PO_{4(aq)} + 3H_2O$

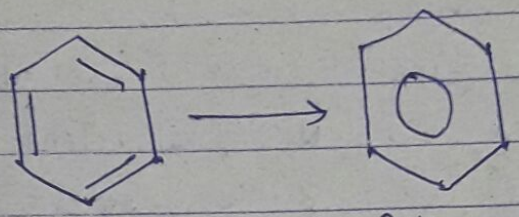
$\Delta H_{ion}(H_3PO_4) = 3\times 57.3 - 150$
$= 21.9$ KJ/mol

(10) Resonance Energy or Resonance Enthalpy : (ΔH_{RE})

→ Resonance means delocalisation of π electrons.

Resonance enthalpy is the enthalpy change when 1 mol of the most stable resonating structure or canonical structure of a compound is converted into resonance hybrid.

→ ΔH_{RE} is -ve (exothermic)

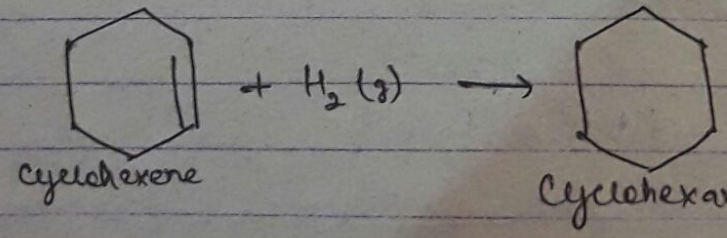
eg. 

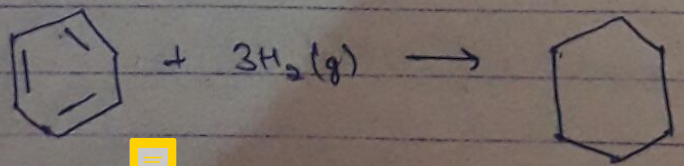
, $\Delta H_r = \Delta H_{RE}$ (C_6H_6)

→ Since, resonating structure is hypothetical (non-existant) the above reaction (for benzene) cannot be used to determine Resonance Energy.

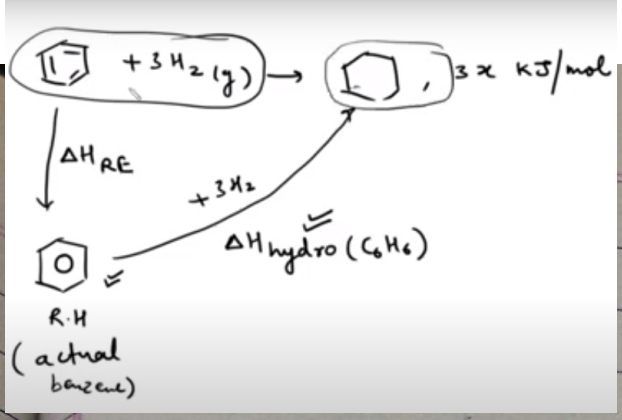
→ Indirectly $\Delta H_{R.E}$ is determined using Bond energy data or enthalpy of combustion data or heat of hydrogenation data.

eg. To determine ΔH_{RE} of Benzene :-





✓ = given



✓ $3x = \Delta H_{RE}(C_6H_6) + \Delta H_{Hydr}(C_6H_6)$ ✓

Can be calculated.

Q. Find heat of combustion of Methyl alcohol CH_3OH at 298K.

	C-H	C-O	O-H	O=O	C=O
B.E (KJ/mol)	414	315.5	464.5	494	711

$\Delta H_{R.E}$ of CO_2 = −143 KJ/mol

$\Delta H_{vap}(CH_3OH(l))$ = 35.5 KJ/mol

$\Delta H_{vap}(H_2O(l))$ = 40.6 KJ/mol.

Soln:

$CH_3OH(l) + 1.5\,O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$, $\Delta H_r = ?$

35.5 | 0 | $-\Delta H_{R.E} = -(-143)$ | 2×40.6

$CH_3OH(g) + 1/2\,O_2(g) \xrightarrow{\Delta H'} O=C=O + 2H_2O(g)$

R.S

applying Hess Law on

$35.5 + 0 + \Delta H' = \Delta H_r + 143 + 2 \times 40.6$ ——— (1)

$\Delta H' = \sum (B.E)_R - \sum (B.E)_P$

$= (3 \times B.E_{C-H} + B.E_{C-O} + B.E_{O-H} + 1.5 \times BE_{O=O}) - (2 \times B.E_{C=O} + 4 \times B.E_{O-H})$

$\boxed{\Delta H_r = -669.7 \text{ KJ/mol}}$

(11) Enthalpy of transition (ΔH_{tran})

It is the enthalpy change when 1 mol of an allotropic form of an element is converted into another allotropic form.

eg. $C(graph) \longrightarrow C(dia)$, $\Delta H = \Delta H_{trans}(C_{graph} \to C_{dia})$
1 mol 1 mol

eg. $S(rhomb) \longrightarrow S(mono)$, $\Delta H_r = \Delta H_{trans}(S_{rhom} \to S_{mono})$

Q. Find enthalpy of transition of following phase change.
$C(dia) \longrightarrow C(graph)$.

$\Delta H_{comb}(dia) = x$ KJ/mol
$\Delta H_{comb}(graph) = y$ KJ/mol

Soln: $\Delta H_r = \Sigma(\Delta H_{comb})_R - \Sigma(\Delta H_{comb})_{prod.}$
$= x - y$ KJ/mol

Imp Note: Difference b/w & Ionisation Energy & Enthalpy of Ionisation. (ΔH_{IE})

By definition: Ionisation Energy is defined at 0 kelvin.
ΔH_{IE} → defined at any temperature (T Kelvin)

$M(g) \longrightarrow M^+(g) + e^-$, (Ionisation process)

Taking all the species ($M(g)$, $M^+(g)$ & e^-) to be monoatomic, using kirchoff's law.

$$\frac{(\Delta H_r)_{T_2} - (\Delta H_r)_{T_1}}{T_2 - T_1} \Rightarrow (\Delta C_p)_r$$

$$\Rightarrow \frac{\Delta H_{IE} - IE}{(T-0)} = \left(\frac{5}{2}R + \frac{5}{2}R\right) - \frac{5}{2}R$$

$M^+(g)$ e^- $M(g)$

$$\Rightarrow \boxed{\Delta H_{IE} = IE + \frac{5}{2}RT}$$ Memorise !

$\Rightarrow$ Similarly, for $X(g) + e^- \rightarrow X^-(g)$

$$\boxed{\Delta H_{eg} = -EA - \frac{5}{2}RT}$$

Electron affinity is defined at '0' kelvin.